1.5 Applications of Matrices: Consistency of System of Linear Equations by Rank Method

In section 1.3.3, we have already defined consistency of a system of linear equation. In this section, we investigate it by using rank method. We state the following theorem without proof:

Theorem 1.14 (Rouché - Capelli Theorem)

A system of linear equations, written in the matrix form as $AX = B$, is consistent if and only if the rank of the coefficient matrix is equal to the rank of the augmented matrix; that is, $\rho(A) = \rho([A \mid B])$.

We apply the theorem in the following examples.

1.5.1 Non-homogeneous Linear Equations

Example 1.29

Test for consistency of the following system of linear equations and if possible solve: $x+2y-z=3,\ 3x-y+2z=1,\ x-2y+3z=3,\ x-y+z+1=0$.

Solution

Here the number of unknowns is 3.

The matrix form of the system is $AX = B$, where

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & -1 & 2 \\ 1 & -2 & 3 \\ 1 & -1 & 1 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 3 \\ 1 \\ 3 \\ -1 \end{bmatrix}.$$

The augmented matrix is $[A \mid B] = \left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 3 & -1 & 2 & 1 \\ 1 & -2 & 3 & 3 \\ 1 & -1 & 1 & -1 \end{array}\right]$.

Applying Gaussian elimination method on $[A \mid B]$, we get

$$[A \mid B] \xrightarrow[R_4 \to R_4 - R_1]{\substack{R_2 \to R_2 - 3R_1, \\ R_3 \to R_3 - R_1,}} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -7 & 5 & -8 \\ 0 & -4 & 4 & 0 \\ 0 & -3 & 2 & -4 \end{array}\right] \xrightarrow[R_4 \to (-1)R_4]{\substack{R_2 \to (-1)R_2, \\ R_3 \to (-1)R_3,}} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 7 & -5 & 8 \\ 0 & 4 & -4 & 0 \\ 0 & 3 & -2 & 4 \end{array}\right]$$

$$\xrightarrow[R_4 \to 7R_4 - 3R_2]{\substack{R_3 \to 7R_3 - 4R_2, \\}} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 7 & -5 & 8 \\ 0 & 0 & -8 & -32 \\ 0 & 0 & 1 & 4 \end{array}\right] \xrightarrow{R_3 \to R_3 \div (-8)} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 7 & -5 & 8 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 1 & 4 \end{array}\right] \xrightarrow{R_4 \to R_4 - R_3} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 7 & -5 & 8 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

There are three non-zero rows in the row-echelon form of $[A \mid B]$. So, $\rho([A \mid B]) = 3$.

So, the row-echelon form of A is $\begin{bmatrix} 1 & 2 & -1 \\ 0 & 7 & -5 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$. There are three non-zero rows in it. So $\rho(A) = 3$.

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